THE REDUCED CLOSURE OF THE IMAGE OF J = (T, S) CARRIES THE RESTRICTED GROUPOID STRUCTURE OVER A PERFECT FIELD

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Allowed standard prerequisites (editorial).

- Definition of the reduced closed subscheme on a closed subset (schemes chapter, Definition schemes-definition-reduced-induced-scheme, canonical tag 01J4; kind definition). cited at statement lines 1930-1932. Defines Z as the reduced closed subscheme on the closure of the image of j.
- The map-groupoids-on-perfect-field-image lemma, of which the groupoid case is a special instance (more-groupoids chapter, Lemma lemma-map-groupoids-on-perfect-field-image, canonical tag 04Q8). cited at proof lines 1943-1945. One route to the result: the ambient groupoid makes the statement a special case.
- A reduced scheme over a perfect field is geometrically reduced (varieties chapter, Lemma varieties-lemma-perfect-reduced, canonical tag 020I). cited at proof lines 1949-1952. Makes Z geometrically reduced over k via either projection, the entry point of the reducedness chain.
- A geometrically reduced scheme stays reduced after any field-extension base change (varieties chapter, Lemma varieties-lemma-geometrically-reduced-any-base-change, canonical tag 035Z). cited at proof lines 1953-1955. Shows Z times_{pr_1, U, pr_0} Z inside Z times Z is reduced, so the set theoretic containment of the field-image lemma is carried by reduced subschemes.
- Set theoretic composition containment for the reduced closure of the image of j over a field (more-groupoids chapter, Lemma lemma-groupoid-on-field-image, canonical tag 04QB; collected on this lane as L22522). cited at proof lines 1956-1959. Provides pr_02(Z times Z) inside Z set theoretically, which by reducedness upgrades to the induced morphism of the restricted fibre product into Z.

Source reference. Original definition 01J4: sources/stacks-chow/tags.txt, source lines 2001–2001. Cited at statement lines 1930-1932: defines Z as the reduced closed subscheme on the closure of the image of j.

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Source reference. Original lemma 04Q8: sources/stacks-chow/tags.txt, source lines 6141–6141. Cited at proof lines 1943-1945: the ambient groupoid makes the statement a special case of this lemma.

Source reference. Original lemma 020I: sources/stacks-chow/tags.txt, source lines 2635–2635. Cited at proof lines 1949-1952: Z is geometrically reduced over the perfect field k via either projection.

Source reference. Original lemma 035Z: sources/stacks-chow/tags.txt, source lines 4146–4146. Cited at proof lines 1953-1955: Z times_{pr_1, U, pr_0} Z inside Z times Z is reduced.

Source reference. Original lemma 04QB: sources/stacks-chow/tags.txt, source lines 6144–6144. Cited at proof lines 1956-1959: the set theoretic containment pr_02(Z times Z) inside Z that reducedness upgrades to the induced morphism.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume U is the spectrum of a perfect field. Let $Z \subset U \times_S U$ be the reduced closed subscheme (see Schemes, Definition 01J4) whose underlying topological space is the closure of the image of $j = (t, s) : R \to U \times_S U$. Then*

$$(U, Z, pr_0|_Z, pr_1|_Z, pr_{02}|_{Z \times_{pr_1, U, pr_0} Z})$$

is a groupoid scheme over S.

Proof. As $(U, U \times_S U, \mathrm{pr}_1, \mathrm{pr}_0, \mathrm{pr}_{02})$ is a groupoid scheme over S this is a special case of Lemma 04Q8. But we can also prove it directly as follows.

We first explain why the statement makes sense. Since U is the spectrum of a perfect field k, the scheme Z is geometrically reduced over k (via either projection), see Varieties, Lemma 020I. Hence the scheme $Z \times_{\mathrm{pr}_1, U, \mathrm{pr}_0} Z \subset Z$ is reduced, see Varieties, Lemma 035Z. Hence by Lemma 04QB we see that pr_{02} induces a morphism $Z \times_{\mathrm{pr}_1, U, \mathrm{pr}_0} Z \to Z$. Finally, it is clear that $\Delta_{U/S}$ factors through Z and that the map $\sigma : U \times_S U \to U \times_S U$, $(x, y) \mapsto (y, x)$ preserves Z. Since $(U, U \times_S U, \mathrm{pr}_0, \mathrm{pr}_1, \mathrm{pr}_{02}, \Delta_{U/S}, \sigma)$ satisfies the axioms of a groupoid, it follows that after restricting to Z they satisfy the axioms. □